Yiran Wang

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Education

- **Yale University** — New Haven, CT
  Master of Science in Biostatistics (Data Science Track) — *Fall 2023 – Summer 2025*
  - **GPA:** 3.70/4.0
  - **Core Modules:** Machine Learning (Biological Data) (H); Machine Learning in Healthcare (H); Advanced Regression Models (HP); Probability Theory (H); Deep Learning (H); Intermediate Machine Learning (H); Statistical Inference (HP); Data Structures; Information Theory

- **Beijing Forestry University** — Beijing, China
  Bachelor of Science in Mathematics and Applied Mathematics — *Fall 2019 – Summer 2023*
  - **GPA:** 3.81/4.0
  - **Core Modules:** Mathematical Analysis (98), Advanced Algebra (Two Semesters: 88, 92), Operations Research (90), Probability Theory and Mathematical Statistics (85, 94), Complex Variables (97), Mathematical Modeling and Experiment (100), Ordinary Differential Equations (97), Stochastic Processes (99), Statistical Software (R Language) (95), Computational Statistics (97), Numerical Computation (100), Algorithm Design and Analysis (98), Multivariate Statistics (85), Abstract Algebra (96), Real Analysis (with measure theory) (100)

Publications

[1] J. Ban$^+$, **Y. Wang**$^+$, B. Liu$^+$, H. Li. "Optimization of venture portfolio based on LSTM and dynamic programming" *AIMS Mathematics*, 2023. [Paper]

[2] **Y. Wang**$^+$, Z. Lu$^+$, Y. Xie$^+$. "Analysis Methodology of C4 olefins by ethanol coupling based on correlation analysis" in *Highlights in Science Engineering and Technology*, 2023. [Paper]

[3] **Y. Wang**, J. Kim. "cNODE: A Continuous-Time Neural ODE Transformer for Early Sepsis Prediction." Accepted for presentation at AMIA Annual Symposium, 2026.

$+$ denotes these authors contributed to the work equally and should be regarded as co-first authors

Experience

- **Yale University, Department of Biomedical Informatics & Data Science** — New Haven, CT
  Research Assistant (Full-Time), with Sai Zhang; previously with Jihoon Kim — *May 2025 - Present*
  - Built person-level and event-level survival models using the All of Us dataset to identify predictors of opioid use disorder risk after first opioid prescription, including time-varying Cox models with assumption checking and model selection.
  - **Project 1: A continuous-time Neural ODE Transformer for early sepsis prediction**
    - Developed cNODE, a missingness-aware continuous-time Neural ODE Transformer for early sepsis prediction from irregular MIMIC-IV ICU time-series data, with explicit physical-time conditioning via additive time embeddings during latent ODE evolution.
    - Built an end-to-end pipeline for cohort extraction, preprocessing, and horizon-specific sequence generation.
    - Evaluated the model across multiple early-warning horizons, benchmarked against baseline methods, and conducted ablation studies on temporal and architectural components.
    - Applied SHAP and latent trajectory analysis to interpret subgroup-specific patient dynamics and improve clinical explainability.
  - **Project 2: Survival Modeling of Opioid Use Disorder Risk in All of Us**
    - Cleaned and harmonized large longitudinal clinical datasets from Google Cloud, did data-processing, feature engineering, and built analysis-ready survival datasets.
    - Developed person-level and event-level Cox proportional hazards models in R using All of Us data to identify predictors of opioid use disorder risk after first opioid prescription, including time-varying exposure modeling.
    - Performed comprehensive model diagnostics and comparative evaluation like Kaplan–Meier curves, Schoenfeld and martingale residual analyses and AIC/BIC, then adapted models to address assumption violations using splines, stratification, and time-varying coefficients.
  - **Project 3(Ongoing): Privacy-Preserving GWAS via Vertical Federated Learning**

- **Yale University, Department of Biostatistics** — New Haven, CT
  Student Research Assistant, advised by Prof. Leying Guan — *Oct. 2023 – May. 2025*
  - **Project: Analysis on Machine Learning Models' Excessive Dependence on Spurious Features.**
    - Investigated model robustness against spurious correlations by distinguishing core and spurious features, and developed SHAP-based analysis to identify minority-group instances.
    - Developed a novel methodology to detect minority-group instances without explicit group labels by analyzing prediction changes under core-feature removal and spurious-feature negation, leveraging ResNet last-layer representations and a Random Forest-based proximity matrix to differentiate spurious and causal information.

- Conducted extensive evaluations on Waterbirds, CelebA, ColorMNIST, and MultiNLI, constructing diverse dataset splits with varying sample balance and class-ratio settings, and benchmarking against JTT, Last-Layer Retraining, and GIC methods.

- **Beijing Forestry University, Department of Applied Mathematics** Beijing, China
  Student Research Assistant, supervised by Dr. Huiru Wang and Dr. Xiaoyu Zhang *Jun. 2022 – May. 2023*
  - **Project 1: Study on the Optimization of Extreme Learning Machine Based on SVM Variants**
    - Evaluated and benchmarked multiple SVM variants, including Twin SVM, SVM-2K, Deep SVM-2K, MvSVM-2C, and MvLDM, focusing on theoretical properties and computational efficiency.
    - Designed and implemented DMvLDM-DNN and DMvLDM-AE, combining SVM-based frameworks with deep neural networks and multi-view consistency constraints to enhance multi-view learning and improve classification robustness.
  - **Project 2: Seedling Growth Analysis Under Salt Stress**
    - Analyzed root and overall plant growth of *Populus euphratica* seedlings under normal and salt-stress conditions using statistical methods.
    - Applied PCA, factor analysis, and multiple linear regression to identify growth traits and predict root growth.

Awards

2022 *National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM)*
2021 *Finalist Award, Mathematical Contest in Modeling (MCM) (top 5%)*

Internship

- **Technical Manager**
  Beijing Forestry University *Aug. 2021 - Dec. 2021*
  - Led the end-to-end development of a WeChat Mini Program enabling mushroom species recognition via photo capture. Tasks involved setting up the web server, configuring the database environment, implementing backend algorithms, devising the APP's interface and functionality and conducting rigorous product testing.

- **Business Analyst**
  Business Unit of Bicycle Service, Didi Chuxing Technology Co.Ltd *Oct. 2022 - Mar. 2023*
  - Designed and analyzed A/B tests for the ride-sharing app's new Safety Toolkit, applying power analysis to determine experiment duration, sample size, and target markets; built a pipeline of evaluation metrics to continuously monitor product health and user experience.
  - Executed cohort analysis and user segmentation using clustering techniques. Devised personalized coupon strategies and bonus calculation algorithms, successfully increasing user retention and maximizing profitability.

Skills

- **Programming**: C/C++, Java, R, Python, SQL, Linux, Slurm, HPC, Git
- **Statistics**: GLMs, Survival analysis, Longitudinal models, Causal inference
- **Machine Learning & Deep Learning**: supervised learning, tree-based methods, regularization, neural networks, transformers, large language models
- **Tools**: MATLAB, SPSS